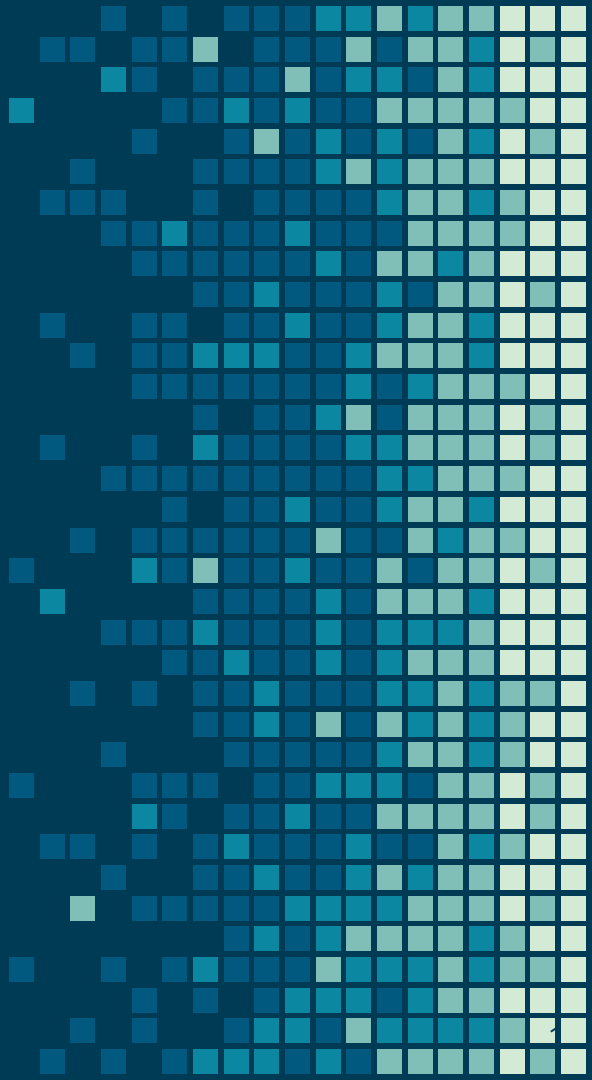
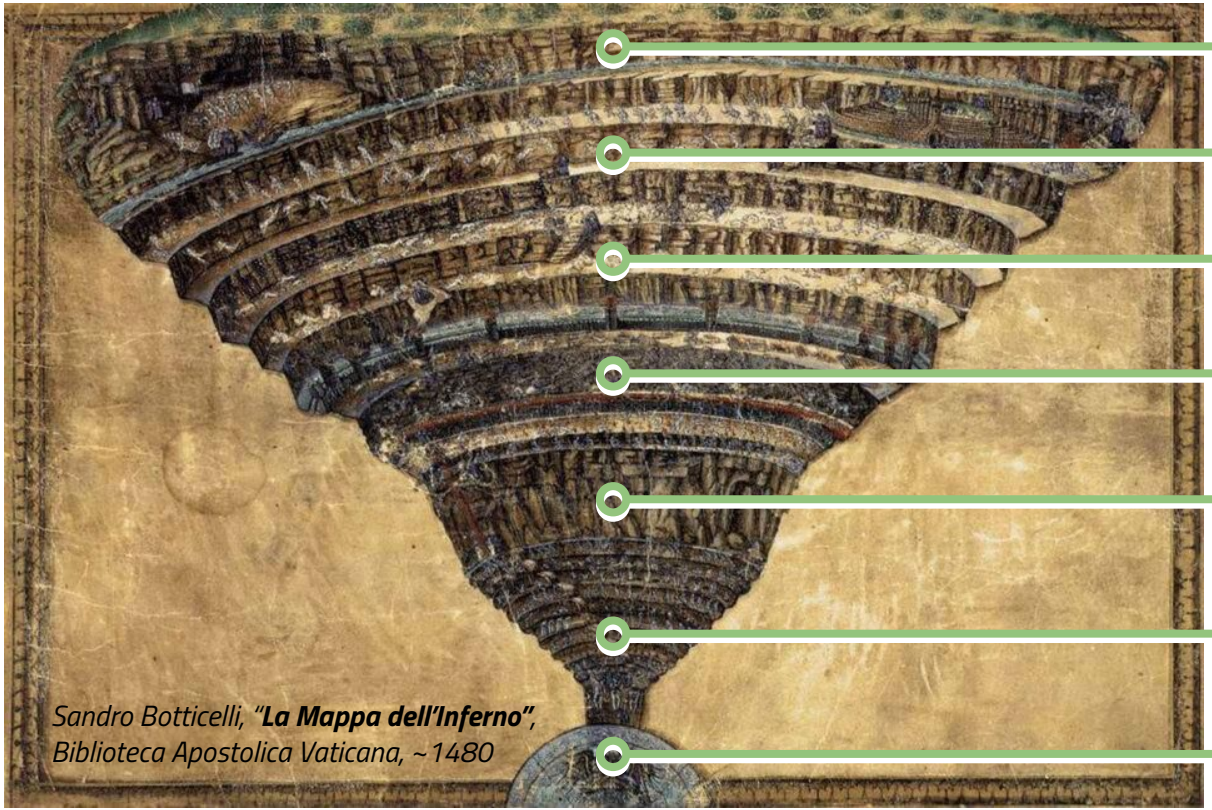
EMBEDDED CONTROL

Prof. Luca De Cicco
luca.decicco@poliba.it
https://c3lab.poliba.it/LDC



GPIOs

Where are we?



*Sandro Botticelli, "**La Mappa dell'Inferno**", Biblioteca Apostolica Vaticana, ~1480*

RTOS

Programming with HAL

Programming with CMSIS

Programming registers

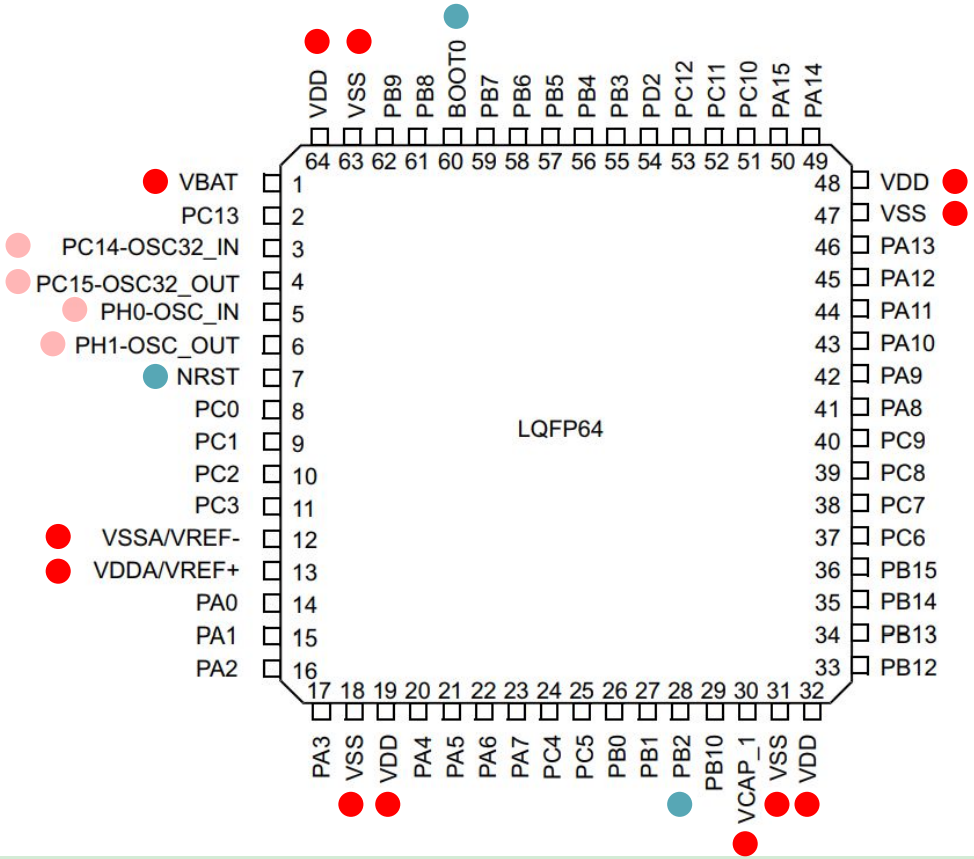
Peripherals Hw

MCU Hw

ARM Cortex-M

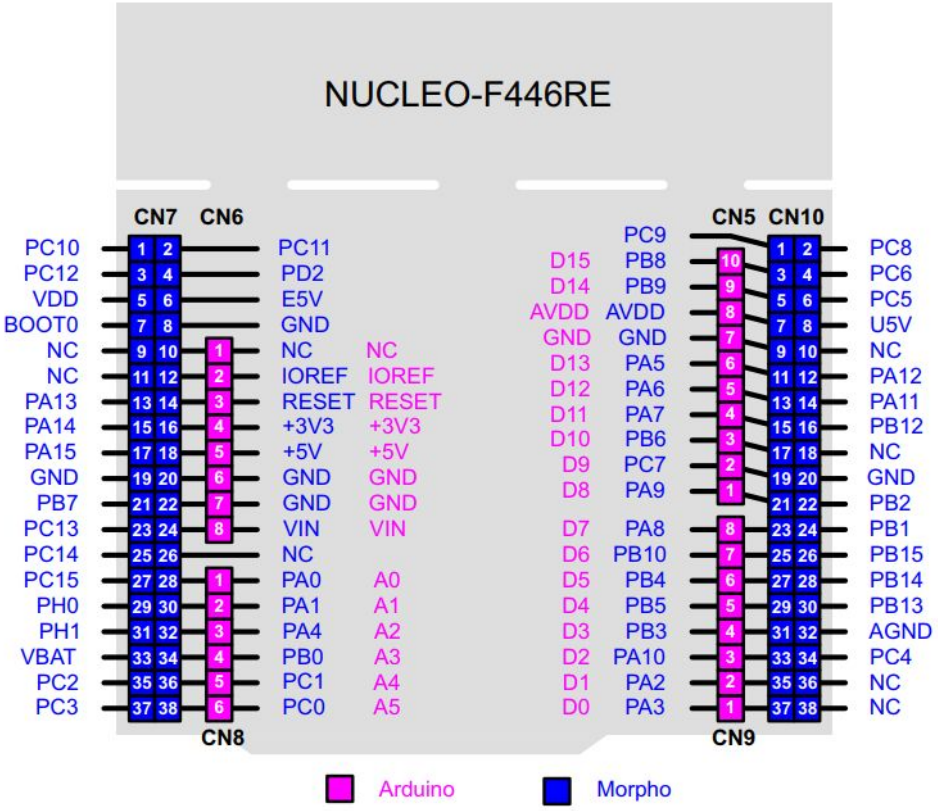
PATH TO ~~REDEMPTION~~ ENLIGHTENMENT

How to connect the MCU to external devs?



Recall all the **Pxy** pins are available to connect external devices to either a specific peripheral or a general purpose I/O

MCU connection to external devs on the Nucleo



General Purpose Input Output (GPIO)

- The number of pins on a MCU is limited (the smaller the size of the MCU package the smaller the pin #)
- Pins that can be configured by software at runtime to perform different functions are called GPIO
- Possible actions:
  - Digital Input that detects if an external voltage is higher/lower than a threshold
  - Digital output that drives the voltage of the pin
  - Analog functions to perform A/D or D/A operations
  - Alternate functions: PWM output, timer-based input capture, USART, SPI, I2C, USB, CAN…

General Purpose Input Output (GPIO)

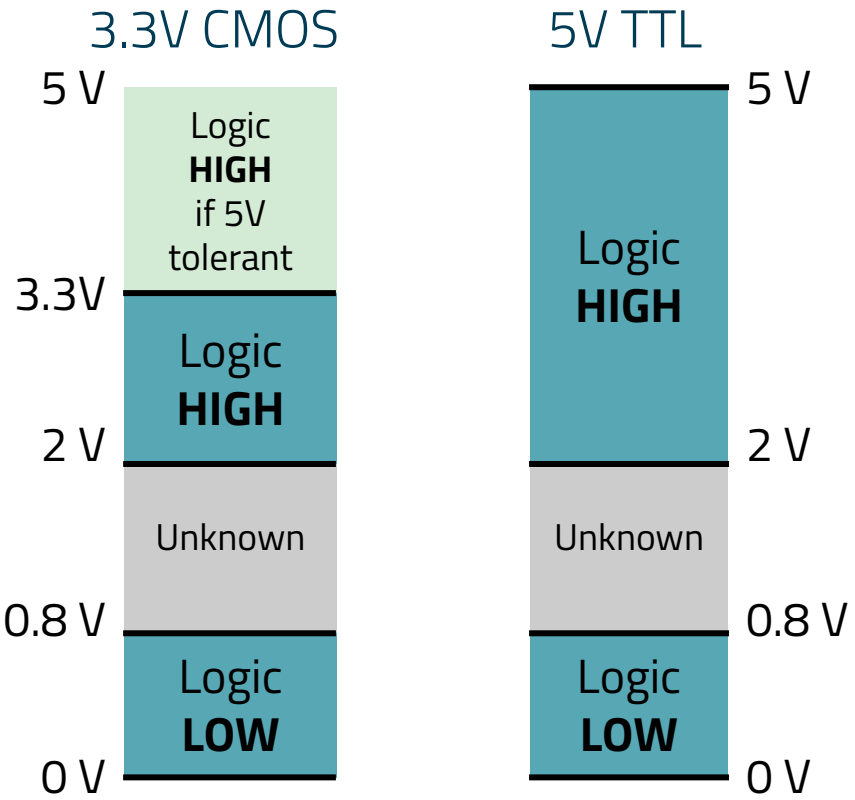
- STM32F446XX associates the GPIOs in **8 ports** called **GPIOA**, **GPIOB**, …, **GPIOH** (note: other MCUs could have a different number of ports)
- Each port is associated to **16 bits**, possibly associated to electric pins of the MCU
- Convention: Pxy, **x** is the port name (A, B,…,H) **y** is the bit (0,…,15). Ex.: pin PC10 is associated to port **C**, bit **10** of the port
- Depending of the MCU package, **not all bits/ports are mapped to electrical pins**. STM32F446RE maps only ports A, B, C to electrical pins (PAy, PBy, PCy)

DIGITAL INPUTS

CMOS Logic Levels

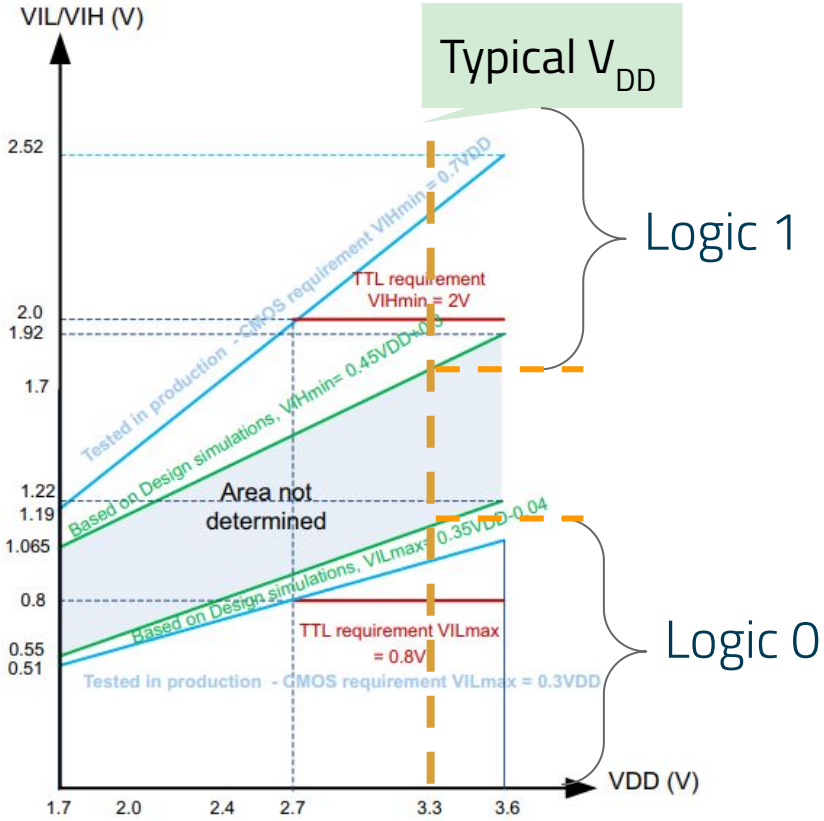
- When we need to read/write a bit of information we need to "read/write" a voltage from/to a GPIO pin
- Definition of LOW voltage (0 bit) HIGH voltage depends on logic family and direction (Input or Output)
- Recall, the maximum voltage we can supply with an STM32 is V_{DD}=3.3V
- However, due to intrinsic additive noise in circuits, we need to be more "elastic" and give some "margins" for the voltage to be considered LOW or HIGH

CMOS (Input) Logic Levels



- Output tolerances are tighter
- It is possible to drive a 5V CMOS from a 3.3V CMOS dev
- STM32 GPIOs are **5V tolerant** i.e., they accept also 5V in input (w/o damaging the ports!)
- **Caution!** In general 5V output connected to a 3.3V input damages the 3.3V port!
- In such cases you need level shifters!

STM32446XX Logic Levels

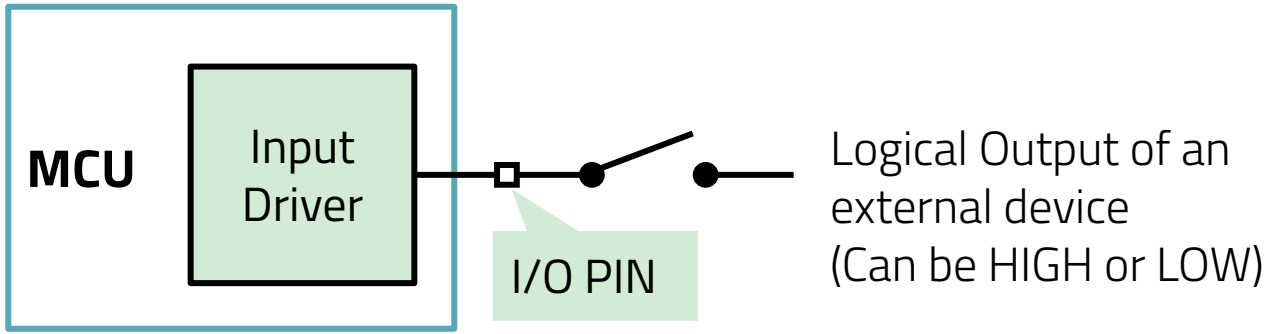


- All I/Os are CMOS and TTL compliant (no software configuration required)
- Their characteristics cover more than the strict CMOS technology or TTL parameters

Source: STM32446xx Datasheet, Figure 31, pg. 120

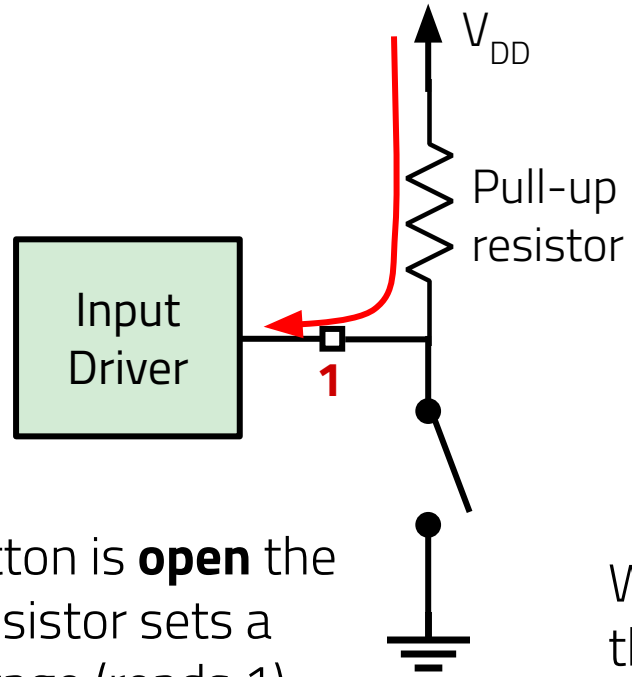
Do not let Digital pins float!

- When using an MCU pin as a digital input it cannot be left open or **floating**

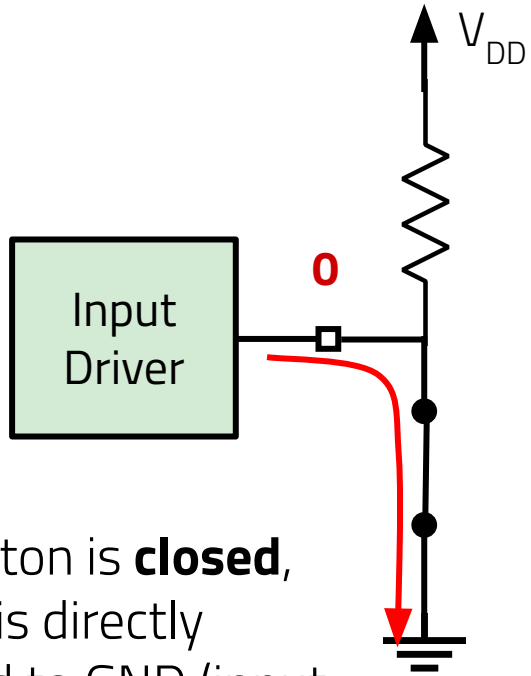


- What's the value of the input when the switch is open?
- The value is **undefined**. (Pins can act like antennas!)
- We have to force a "*default state*" when the switch is open

Pull-up resistor (default: HIGH)

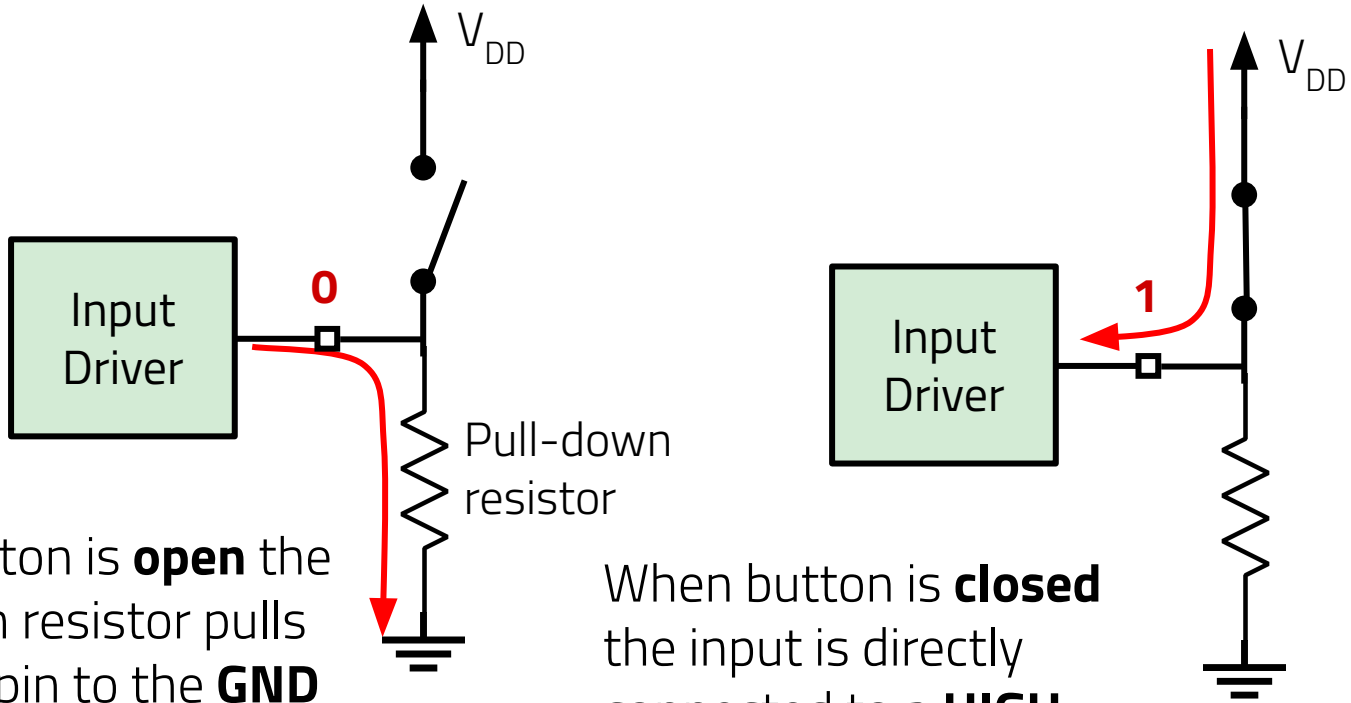


When button is **open** the pull-up resistor sets a **HIGH** voltage (reads 1) on the input pin



When button is **closed**, the input is directly connected to GND (input reads 0)

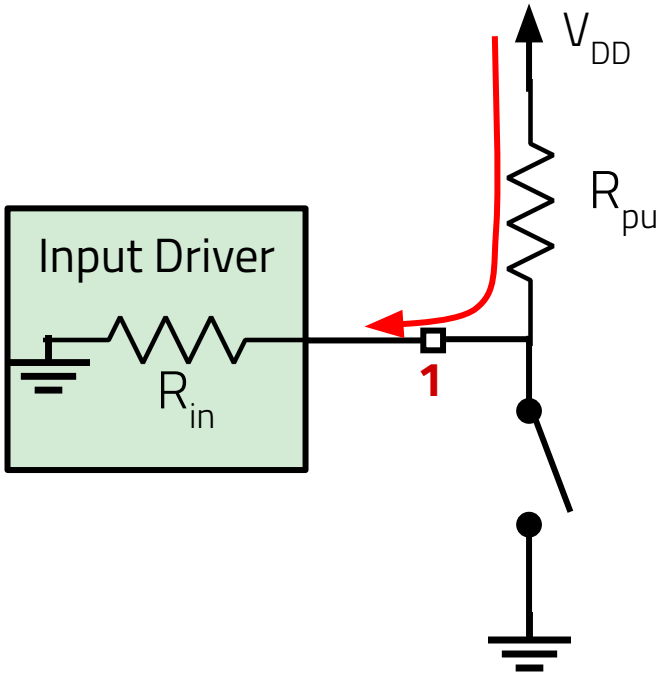
Pull-down resistor (default: LOW)



When button is **open** the pull-down resistor pulls the input pin to the **GND** (reads 0)

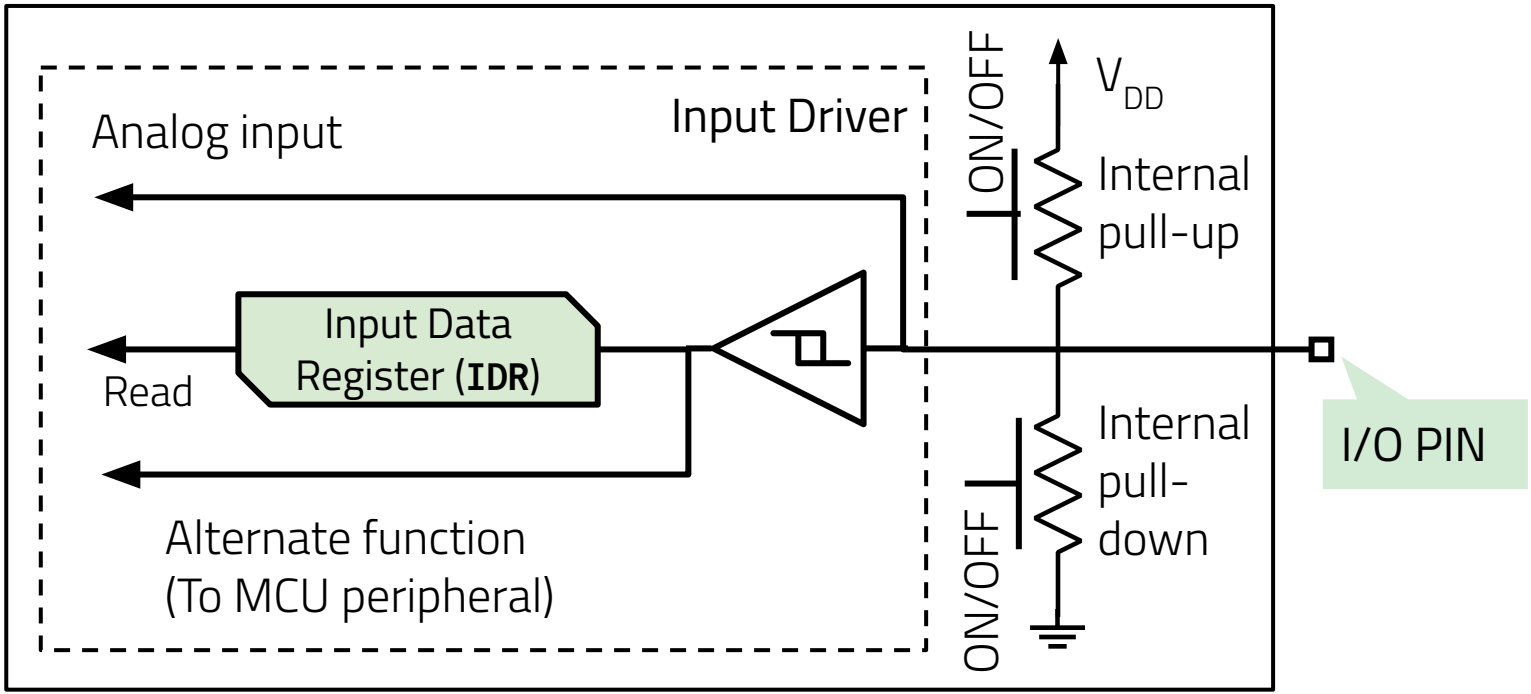
When button is **closed** the input is directly connected to a **HIGH** voltage (reads 1)

How to size pull-up/pull-down resistors?



- A value too low (f.i. zero) would not limit the current drained by the MCU. Issues: power dissipation, damaging the pin!
- Weak pull-up: A too large value makes the current drained very low, but could cause a large voltage drop across the resistor. The read voltage could be too low!
- Typical good values in the [1-10]kΩ range (4.7kΩ is the typical used value)
- Also: large pull-ups make the transient slow when the logic value changes. **Why?**

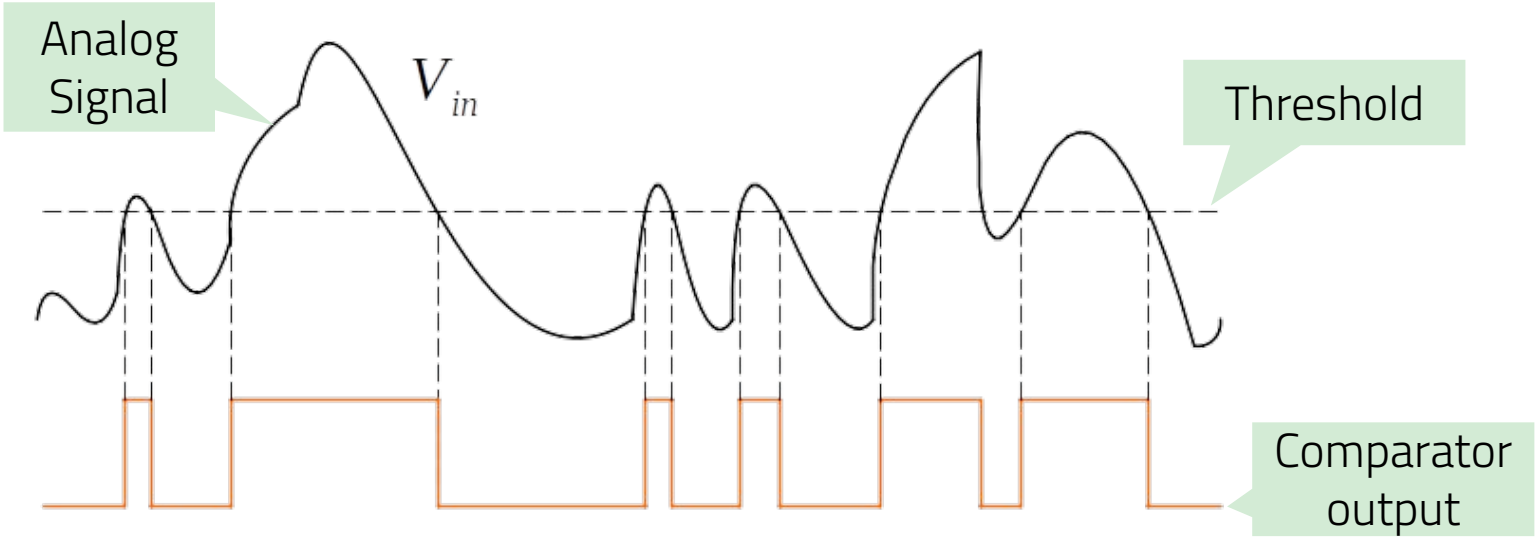
STM32446xx Input Driver



Internal pull-up and pull-down (40kΩ) can be turned ON/OFF through registers

Source: STM32F446re datasheet, pag. 117, Table 56 " I/O static characteristics"

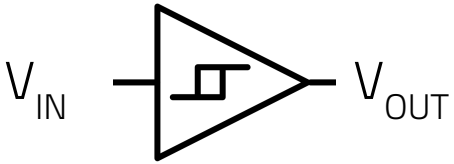
Comparator vs Schmitt Trigger



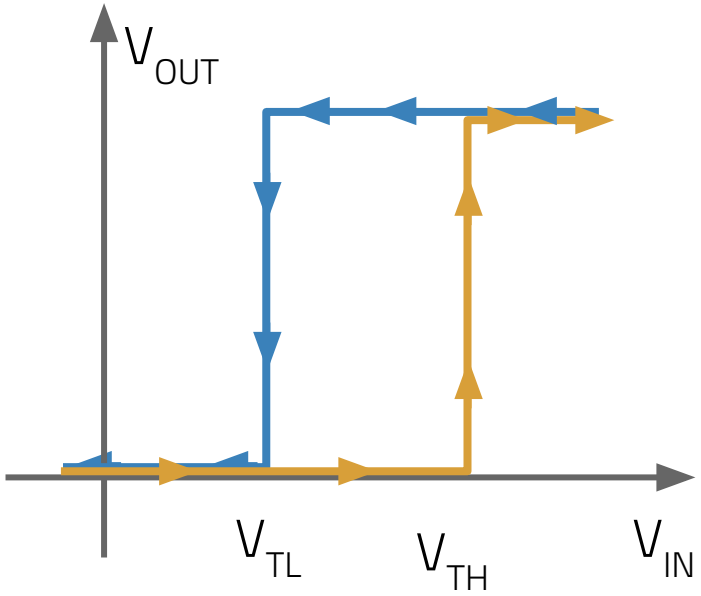
Whenever the analog voltage is higher than a threshold, a digital HIGH is obtained (and vice-versa)

Drawback: poor noise immunity

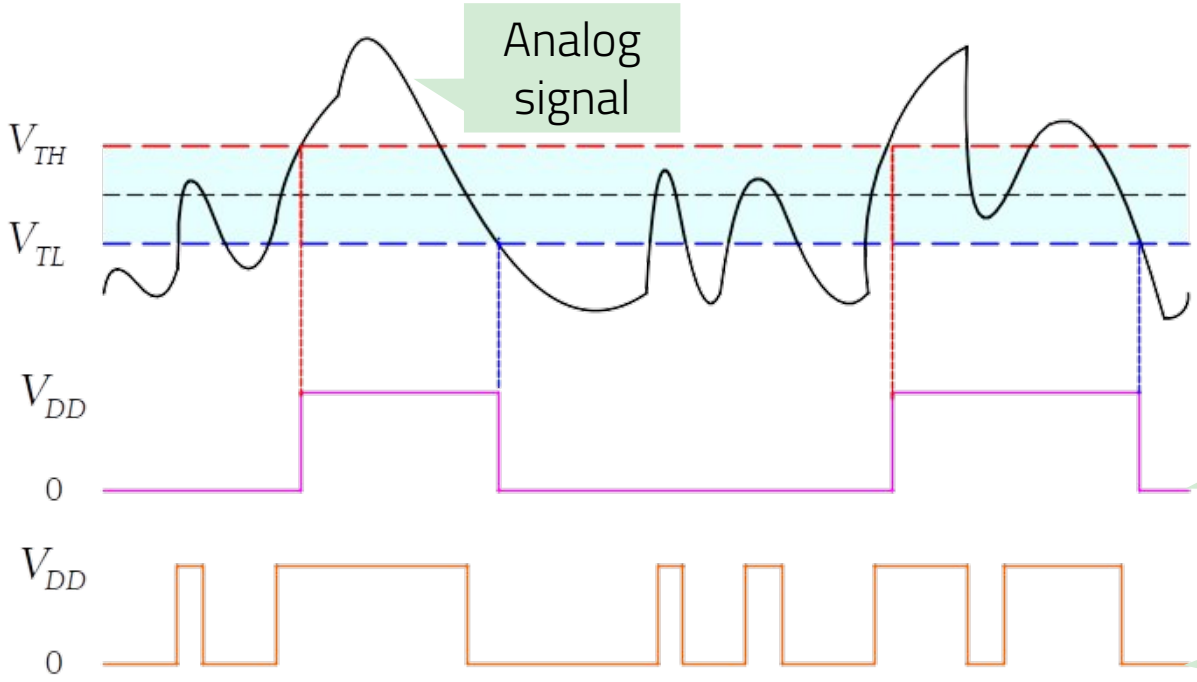
Comparator vs **Schmitt Trigger**



Circuit implementing some sort of “two states FSM”:
Triggers to HIGH when $V_{IN}>V_{TH}$
Triggers to LOW when $V_{IN}<V_{TL}$
If $V_{TL}<V_{IN}<V_{TH}$, V_{out} does not change
state



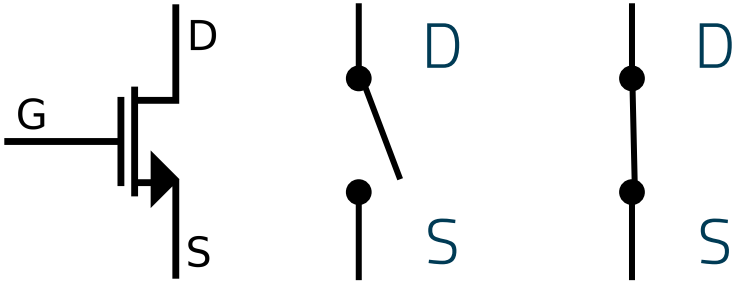
Comparator vs **Schmitt Trigger**



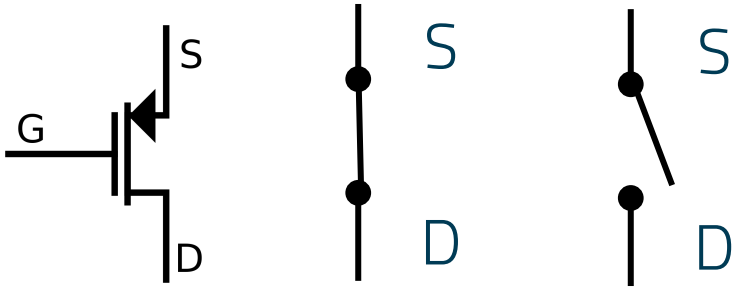
Better noise immunity with respect to the comparator

DIGITAL OUTPUTS

MOSFET (N-MOS, P-MOS) as a switch



N-MOS $V_{GS}<V_{SAT}$ $V_{GS}>V_{SAT}$

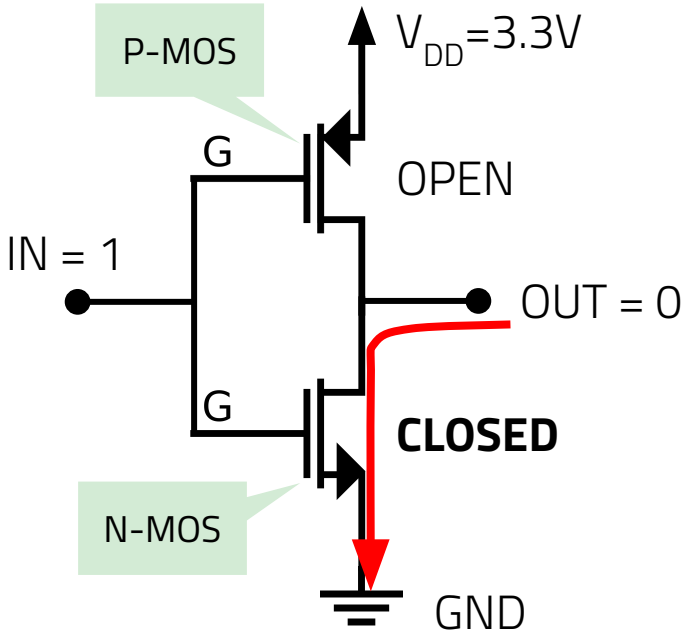


P-MOS $V_{GS}<-V_{SAT}$ $V_{GS}>-V_{SAT}$

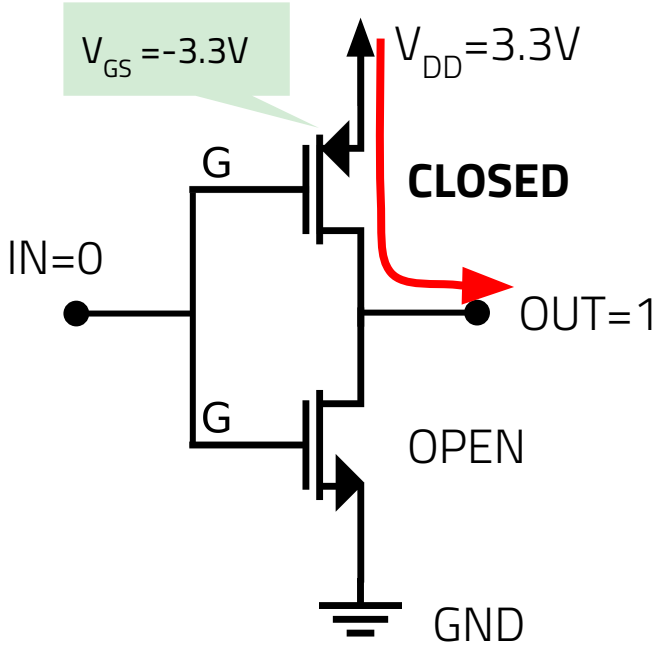
N-MOS act as a switch that is
OPEN (highZ) : if V_{GS} is low
CLOSED : if V_{GS} is high

P-MOS is the inverse

Totem Pole/Push-Pull Configuration (MOS)

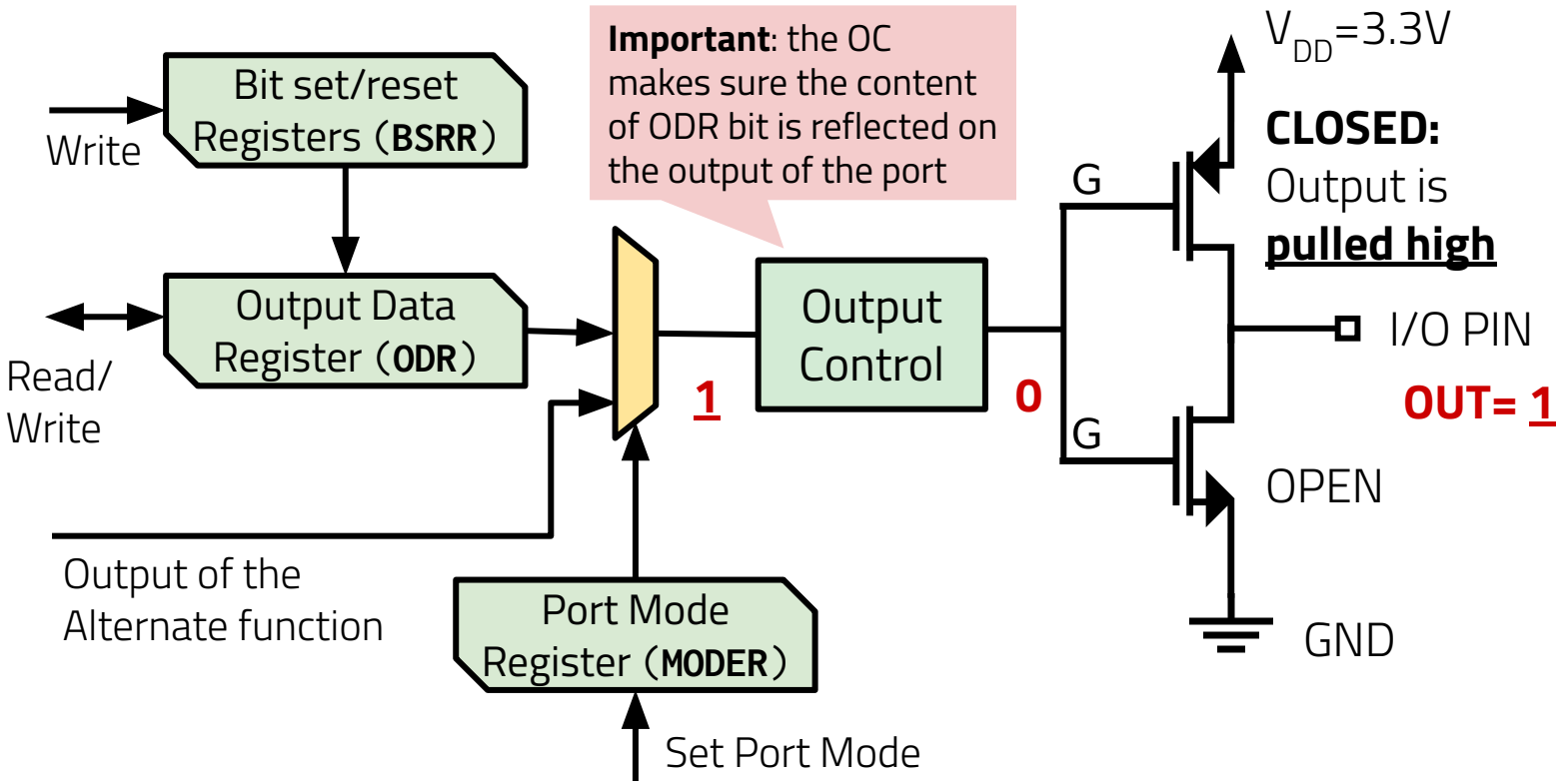


Current Sink



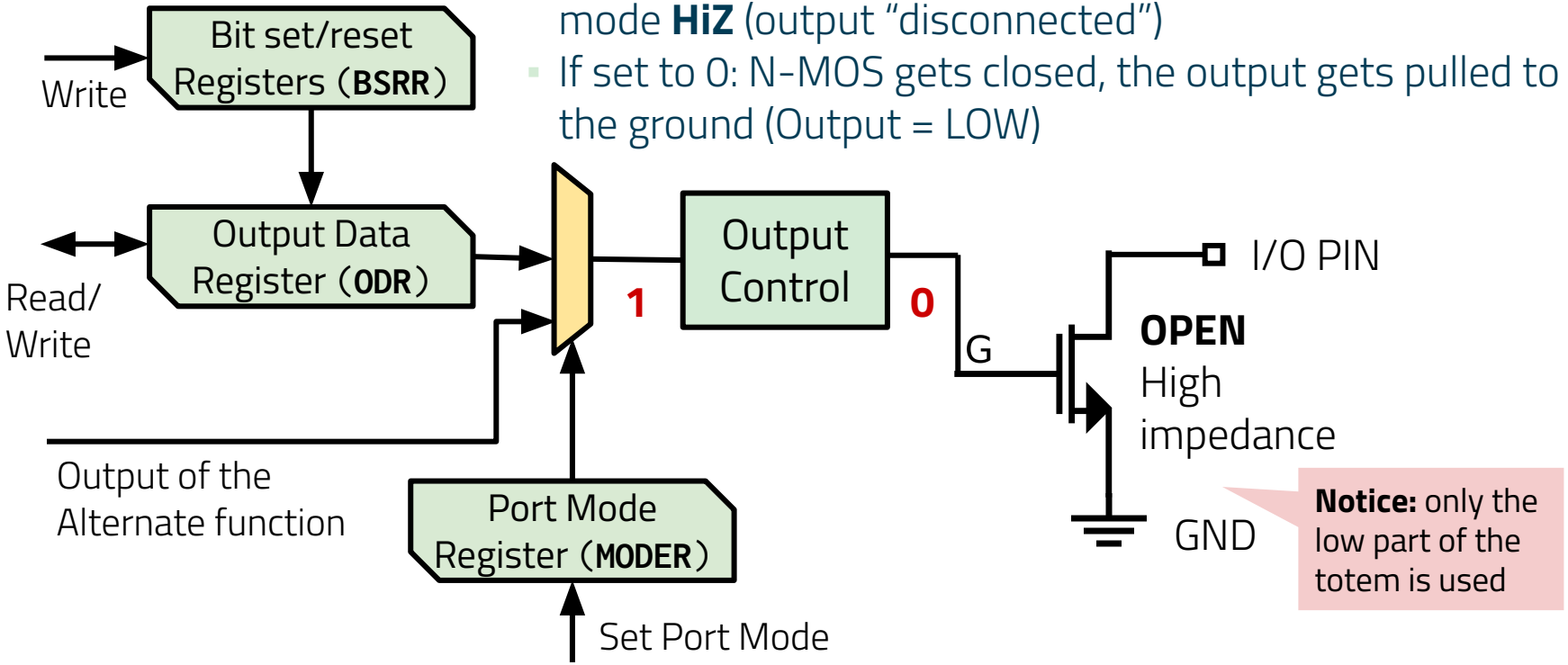
Current Source

Simplified MCU **Output** Stage Push/Pull

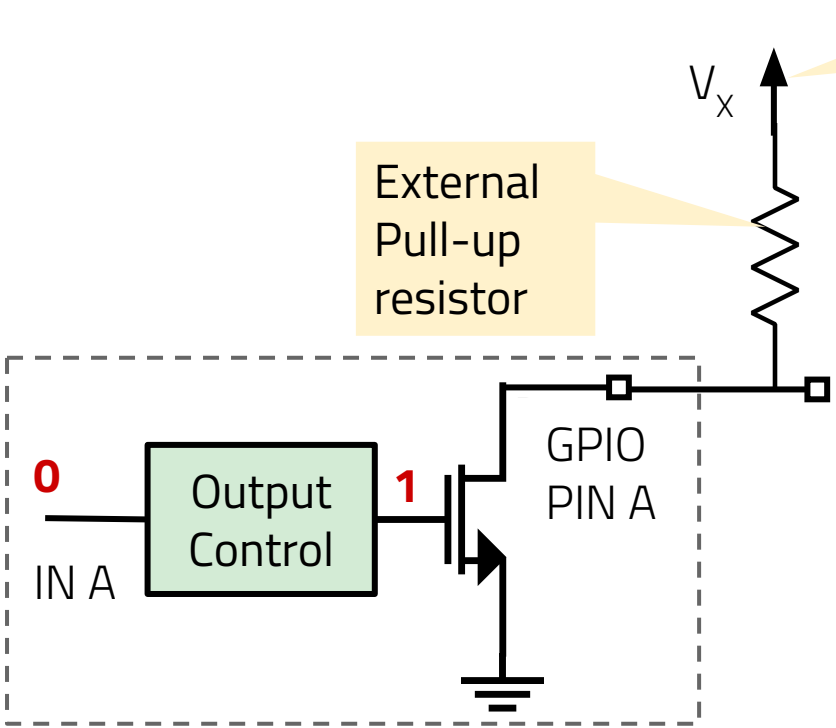


Simplified MCU **Output** Stage **Open Drain**

- If set to 1: N-MOS is open, the output is in high impedance mode **HiZ** (output "disconnected")
- If set to 0: N-MOS gets closed, the output gets pulled to the ground (Output = LOW)

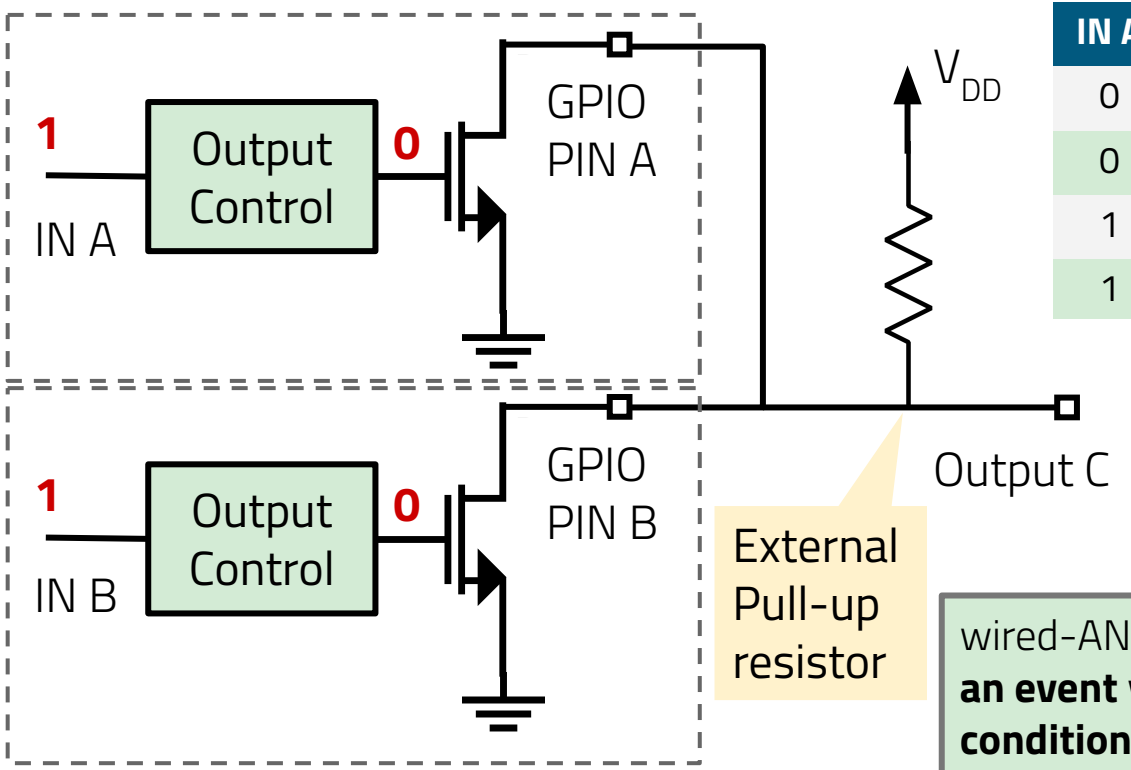


Open-drain usage



- When a GPIO is used in open-drain the output can be used to pull-up a pin to any voltage!
- This feature can be used when a GPIO pin is used as input to another system requiring a higher level of input voltage

Open-drain port usage: wired-AND



IN A	IN B	Out A	Out B	Out C
0	0	GND	GND	0
0	1	GND	HiZ	0
1	0	HiZ	GND	0
1	1	HiZ	HiZ	1

Can be generalized to a **wired-AND of N inputs**

wired-AND can be used to **generate an event** when a number of **conditions hold simultaneously**

References

[AN] AN4899, Application note “STM32 GPIO configuration for hardware settings and low-power consumption”
[DS] STM32F446re datasheet, pag. 119, Table 56

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